**General Instructions:**

**Read the following instructions carefully and follow them:**

1. This question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into 5 sections A, B, C, D and E.
3. In Section A, Question numbers 1-18 are multiple choice questions (MCQs) and question no.19 and 20 are Assertion- Reason based questions of 1 mark each.
4. In Section B, Question numbers 21-25 are very short answer (VSA) type questions, carrying 02 marks each.
5. In Section C, Question numbers 26-31 are short answer (SA) type questions, carrying 03 marks each.
6. In Section D, Question numbers 32-35 are long answer (LA) type questions, carrying 05 marks each.
7. In Section E, Question numbers 36-38 are case study-based questions carrying 4 marks each with sub parts of the values of 1, 1 and 2 marks each respectively.
8. There is no overall choice. However, an internal choice in 2 questions of Section B, 2 questions of Section C and 2 questions of Section D has been provided. An internal choice has been provided in all the 2 marks questions of Section E.
9. Draw neat and clean figures wherever required. Take $\pi = \frac{22}{7}$ wherever required if not stated.
10. Use of calculators is not allowed.

## SECTION - A

### (Multiple Choice Questions)

Each MCQ of 1mark, has four options with only one correct option, choose the correct option

**Q1.** The value of $0.\overline{2}$ in the form $\frac{p}{q}$, where $p$ and $q$ are integers and $q \neq 0$, is **(1)**

(a) $\frac{1}{5}$ (b) $\frac{2}{9}$

(c) $\frac{2}{5}$ (d) $\frac{1}{8}$

**Q2.** The coefficient of $x$ in the expansion of $(x-3)^3$ is **(1)**

(a) $-9$ (b) 9 (c) $-27$ (d) 27

Q3. The points $(2, -1)$, $(6, -5)$ and $(-3, -2)$ (1)

(a) lie in the $II^{nd}$ quadrant (b) lie in the $III^{rd}$ quadrant (c) lie in the $IV^{th}$ quadrant (d) do not lie in the same quadrant

Q4. Which of the following points lies on the line $y = x$? (1)

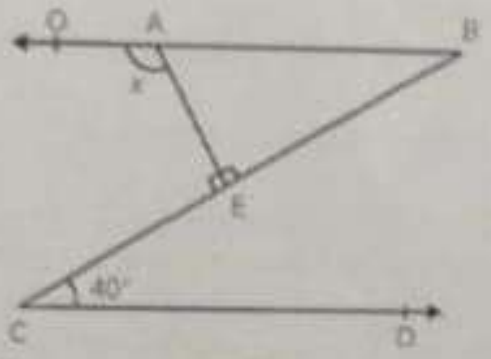
(a) $(a, a)$ (b) $(0, a)$ (c) $(a, 0)$ (d) $(a, -a)$

Q5. Euclid stated that all right angles are equal to each other in the form of (1)

(a) an axiom (b) a definition

(c) a postulate (d) a proof

Q6. In the adjoining figure, if OB ∥ CD, $\angle BCD = 40°$ and AE ⊥ BC then $\angle OAE$ is equal to (1)




(a) 110° (b) 135° (c) 130° (d) 115°

Q7. Ritesh wants to prove that $\Delta FGH \cong \Delta JKL$ using SAS rule. He knows that FG = JK and FH = JL. The additional piece of information that he needs is (1)

(a) $\angle F = \angle J$ (b) $\angle H = \angle L$ (c) $\angle G = \angle K$ (d) $\angle F = \angle G$

Q8. Two right circular cones of equal curved surface areas have slant heights in the ratio of 3 : 5. The ratio of their radii is (1)

(a) 4 : 1 (b) 3 : 5 (c) 5 : 3 (d) 4 : 5

Q9. Zero of the zero polynomial is (1)

(a) 0 (b) 1 (c) any real number (d) not defined

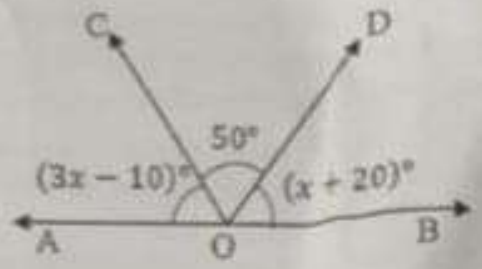
Q10. The value of $\frac{\sqrt{32} + \sqrt{48}}{\sqrt{8} + \sqrt{12}}$ is equal to (1)

(a) $\sqrt{2}$ (b) 2 (c) 4 (d) 8

**Q11.** The volume of sphere of radius $2r$ is **(1)**

(a) $\frac{32\pi r^3}{3}$ (b) $\frac{16\pi r^3}{3}$ (c) $\frac{8\pi r^3}{3}$ (d) $\frac{64\pi r^3}{3}$

**Q12.** In the adjoining figure, AOB is a straight line. If $\angle AOC = (3x - 10)^\circ$, $\angle COD = 50^\circ$ and $\angle BOD = (x + 20)^\circ$ then find $\angle AOC$. **(1)**




(a) 40° (b) 60° (c) 80° (d) 50°

**Q13.** The edges of a triangular board are 6 cm, 8 cm and 10 cm. The cost of painting it at the rate of 9 paise per $cm^2$ is **(1)**

(a) ₹2.00 (b) ₹2.16 (c) ₹2.48 (d) ₹3.00

**Q14.** In a frequency distribution, the mid value of a class is 10 and the width of the class is 6. The lower limit of the class is **(1)**

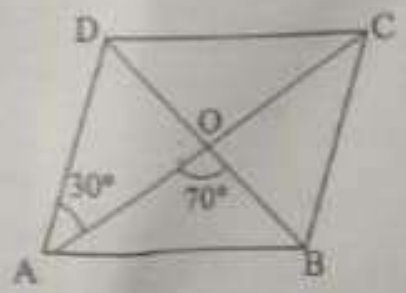
(a) 6 (b) 7 (c) 8 (d) 12

**Q15.** Which of the below options represents the equation $5y - 8x = 7(x + y) - 9$ in the standard form $ax + by + c = 0$? **(1)**

(a) $-x + 6y - 9 = 0$ (b) $-x + 12y - 9 = 0$

(c) $15x + 2y - 9 = 0$ (d) $-15x - 2y - 9 = 0$

**Q16.** The diagonals AC and BD of a parallelogram ABCD intersect each other at the point O such that $\angle DAC = 30^\circ$ and $\angle AOB = 70^\circ$. Then $\angle DBC$ is **(1)**




(a) 50° (b) 40° (c) 60° (d) 70°

**Q17.** The radius of a hemispherical balloon increases from 6 cm to 12 cm as air is being pumped into it. The ratios of the surface areas of the balloon in the two cases is **(1)**

(a) 1:4 (b) 1:3 (c) 2:3 (d) 2:1

Q18. Two quantities P and Q are such that P = Q. Which of these equations illustrate the Euclid's axiom, "If equals are added to equals, the wholes are equals"? (1)

(a) $P + x = Q - x$ (b) $P + x = Q + x$ (c) $P + x = Q$ (d) $P \times x = Q$

**Each of the following questions contains two statements i.e., ASSERTION and REASON, and has following four choices. Only one of which is the correct answer.**

Q19. **Assertion (A)**: If the diagonals of a parallelogram are equal, then it is a rectangle. (1)

**Reason (R)**: The diagonals of parallelogram bisect each other at right angles.

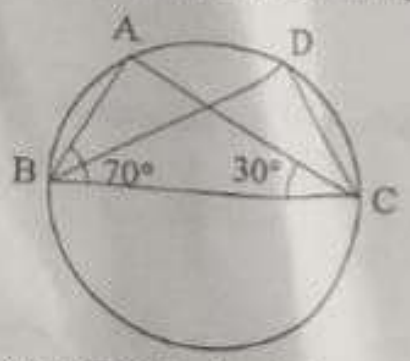
(a) Both assertion (A) and reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(b) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).

(c) Assertions (A) is true but reason (R) is false.

(d) Assertions (A) is false but reason (R) is true.

Q20. **Assertion (A):** In the given figure, if $\angle ABC = 70°$ and $\angle ACB = 30°$, then $\angle BDC = 80°$. (1)




**Reason(R):** Angles in the same segment of a circle are equal.

(a) Both assertion (A) and reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(b) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A)

(c) Assertions (A) is true but reason (R) is false.

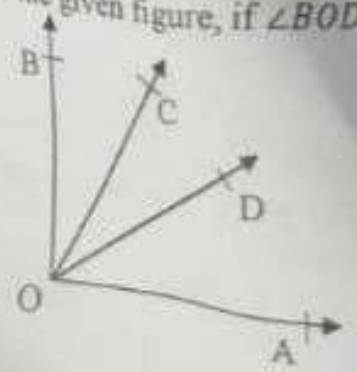
(d) Assertions (A) is false but reason (R) is true.

## Section - B

### (Very Short Answers)

This section comprises of VSA of 2 marks each

Q21. In the given figure, if $\angle BOD = \angle AOC$, then $\angle BOC = \angle AOD$. (2)



**Q22.** Find the value of $k$, if $x = 2k - 3$ and $y = k + 2$ is a solution of the equation $2x + 3y + 15 = 0$. **(2)**

**Q23.** Find the volume of the sphere whose surface area is 154 $cm^2$. **(2)**

**Q24.** **(A)** Find the value of $a$ if $(x + 1)$ is a factor of $a^2x^3 + 5ax + 5a + 4$. **(2)**

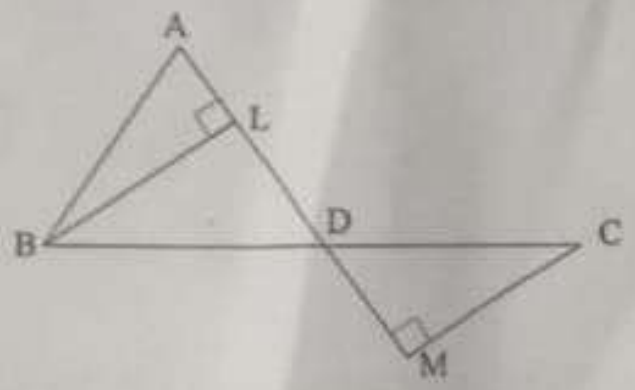
OR

**(B)** Find the zeroes of the polynomial $p(x) = (x - 2)^2 - (x + 2)^2$.

**Q25.** **(A)** If the altitudes from two vertices of a triangle to opposite sides are equal, prove that the triangle is isosceles. **(2)**

**OR**

**(B)** AD is a median and BL and CM are perpendiculars drawn from B and C respectively on AD and AD produced. Prove that BL = CM.




## Section - C

### (Short Answers)

This section comprises of SA type questions of 3 marks each

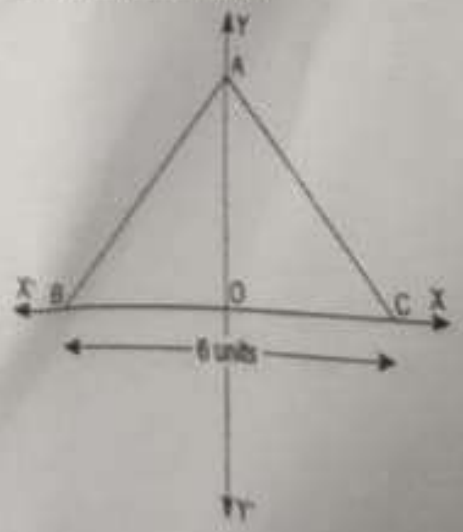
**Q26.** **(A)** Point A is chosen on y-axis in such a way that $\Delta ABC$ is an equilateral triangle. The base BC = 6 units. **(3)**

Find the coordinates of

(i) the mid-point of BC

(ii) the length of AO.

(iii) the vertices of a triangle.



**OR**

**(B)** Find the coordinates of the point:

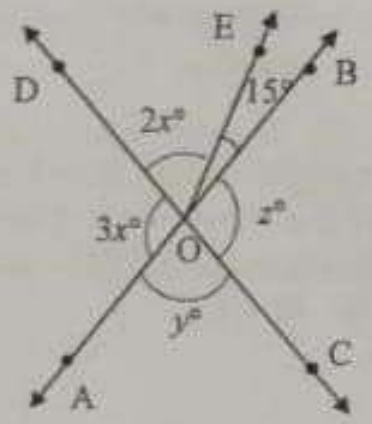
(i) Whose perpendicular distance from the y-axis, measured along the negative direction of $x$-axis, is 4 units, and the perpendicular distance from the $x$-axis, measured along the positive direction of y-axis, is 3 units.

(ii) Whose abscissa is $-6$, lies in the III$^{rd}$ quadrant and perpendicular distance from the $x$-axis is 4 units.

(iii) Whose ordinate is 8 and which lies on the $y$-axis.

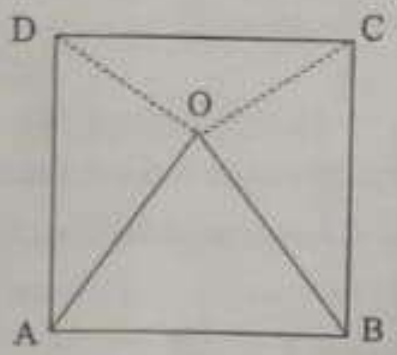
**Q27.** Simplify $(2\sqrt{2}-5)^2+(3\sqrt{2}+\sqrt{3})^2-(\sqrt{2}-1)^2$. **(3)**

**Q28.** In the adjoining figure, two lines AB and CD intersect at O, If $\angle BOE = 15°$, find the values of $x, y$ and $z$. **(3)**




**Q29.** If $x+y+z=8$ and $xy+yz+zx=20$, find the value of $x^3+y^3+z^3-3xyz$. **(3)**

**Q30.** O is a point in the interior of a square ABCD such that OAB is an equilateral triangle. Show that $\Delta OCD$ is an isosceles triangle. **(3)**




**Q31.** **(A)** How many metres of cloth, 2.5 m wide, will be required to make a conical tent whose base radius is 7 m and height 24 m? **(3)**

**OR**

**(B)** A dome of a building is in the form of a hemisphere. From inside, it was white washed at the cost of ₹1232. If the cost of white washing is 400 paisa per square meter, find the volume of the dome.

## Section - D
### (Long Answers)

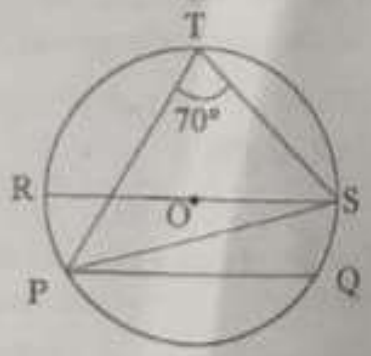
This section comprises of LA type questions of 5 marks each

**Q32.** **(A)** If $\left(\frac{3}{5}\right)^{2x} \times \left(\frac{5}{3}\right)^{\frac{4x}{3}} = \frac{81}{625}$, find the value of $x$. **(5)**

**OR**

**(B)** If $a = \frac{\sqrt{5}+\sqrt{10}}{\sqrt{10}-\sqrt{5}}$ and $b = \frac{\sqrt{10}-\sqrt{5}}{\sqrt{10}+\sqrt{5}}$, rationalise the denominators of $a$ and $b$. Hence show that $\sqrt{a} - \sqrt{b} = 2$.

**Q33.** **(A)** In the adjoining figure RS is a diameter of a circle with centre O, and T, P are points on the circle such that $\angle PTS = 70°$. If the chord PQ is parallel to the diameter RS, find $\angle SPQ$ and $\angle PRS$. **(5)**




**OR**

**(B)** Prove that the quadrilateral formed (if possible) by the internal angle bisectors of any quadrilateral is cyclic.

**Q34.** The point A($b$, 0) satisfies the linear equation $y = 8x + 3$. **(5)**

(i) Find the value of $b$.

(ii) Check if (2, 18) and (−1, −5) are solutions of $y = 8x + 3$.

(iii) Find two more solutions of $y = 8x + 3$.

**Q35.** The given frequency table shows the rate at which the heart beats of an athlete running on a treadmill at a constant speed. **(5)**

| Time (in seconds) | 0 - 60 | 60 - 120 | 120 - 180 | 180 - 240 | 240 - 300 |
|---|---|---|---|---|---|
| Heart beat rate | 85 | 100 | 120 | 110 | 110 |

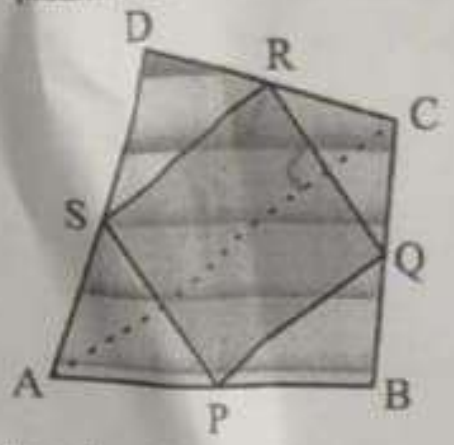
Draw a frequency polygon and histogram.

## Section E

### (Case-study Based Questions)

This section comprises of 3 case-study based questions of 4 marks each with three sub-parts.

**Q36.** Maths teacher of class IX formed a parallelogram PQRS by joining the mid-points of a quadrilateral ABCD as shown in the below figure such that SR ∥ AC ∥ PQ.




Based on the above information, answer the following questions.

(i) If AC = 10 cm, find the lengths of SR and PQ. **(1)**

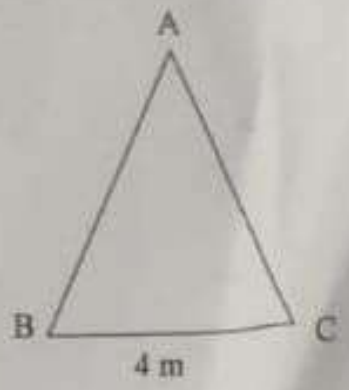
(ii) If SP = 3 cm, find RQ. **(1)**

(iii)A If $\angle RSP = 80°$, find the measure of $\angle RQP$ and $\angle SRQ$. **(2)**

**OR**

(iii)B Find the measure of $\angle R$ and $\angle S$ if $\angle P : \angle Q = 1 : 4$

7. Triangles are used in bridges because they evenly distribute weight without changing their proportions. When force is applied on a shape like a rectangle it would flatten out. Before triangles were used in bridges, they were weak and could not be very big. To solve that problem engineers would put a post in the middle of a square and make it more sturdy. Isosceles triangles were used to construct a bridge in which the base (unequal side) of an isosceles triangle is 4 m and its perimeter is 20 m. One of such triangle is shown in the figure below.




(i) Find the length of equal sides of . (1)

(ii) Find the cost of covering the border of one isosceles triangle at the rate of ₹200 per metre. (1)

(iii)A Find the area of the given isosceles triangle ABC. (2)

**OR**

(iii)B If the sides of a triangle are in the ratio 3 : 5 : 7 and its perimeter is 300 m, find the actual lengths of the triangle and the semi perimeter.

Q38. A small startup's weekly profit (in ₹) is modelled by $p(x) = -x^3 + 9x^2 - 23x + 15$, where $x$ is the number of units produced (in hundreds). Based on the above information, answer the following questions:

(i) Find the profit (or loss) if 0 units are produced. (1)

(ii) Use the Factor Theorem to show that $(x - 1)$ is a factor of the given polynomial. (1)

(iii)A The profit polynomial can be written as $p(x) = -(x - 1)(x^2 - 8x + 15)$. Find the other two values of $x$ (apart from $x = 1$) where the profit polynomial becomes zero. (2)

**OR**

(iii)B A manager uses a different model to calculate combined expenses for three departments of the startup using the expression $(x - 2y + 3z)^2$. Expand this expression using a suitable algebraic identity.